

ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-ADDMOTOR (based on PROJECT-CE)

ORIGINATION: Prof Kidd, UMBC

PURPOSE OF CHANGE: Add motor fan

DOCUMENT NUMBER: ECN2

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EXECUTIVE SUMMARY:

This ECN documents the changes required to extend the PROJECT-CE Arduino LED controller to include a DC motor-driven fan. The customer has added requirements specifying a fan with four speed settings (OFF, LOW, MEDIUM, HIGH) controlled by a momentary pushbutton. The fan must be driven by a PWM signal through an N-channel MOSFET circuit including a flyback diode and smoothing capacitor. The round-robin cyclic executive scheduler from the previous project has been extended with a new fan task to ensure concurrent operation without interference to the existing LED and serial monitor functionality. New testing requirements T.6 through T.12 have been added to validate the fan behavior.

HIGH-LEVEL DESCRIPTION:

Add a PWM-controlled DC motor fan with a momentary pushbutton speed controller to the existing Arduino LED blink-rate system.

REASON FOR CHANGE:

The customer has added new requirements (C6–C12) requesting a motor-driven fan with variable speed control. These requirements necessitate hardware additions (MOSFET, diode, capacitor, button, motor) and software additions (Task_Fan in the round-robin scheduler).

CHANGED DOCUMENTS:

PROJECT REPORT (Demory.cmpe311.spring26.project#2):

- Customer Requirements: Added C6, C7, C8, C9, C10, C11, C12 (4/14/2026, Demory)
- High-Level Requirements: Added HL.7 through HL.18 (4/14/2026, Demory)

- Testing Requirements: Added T.6 through T.12 (4/14/2026, Demory)
- Design Pre-requisites: Added motor, MOSFET, diode, capacitor, button, 9V battery, resistors (4/14/2026, Demory)
- Testing table: Added rows T6–T12 (4/14/2026, Demory)

ADDED DOCUMENTS:

None —

REMOVED DOCUMENTS:

None.

REPLACED DOCUMENTS:

Project Report project#1 replaced by project#3 (Demory.cmpe311.spring26.project#2) incorporating all additions from this ECN.

ANY ADDITIONAL DESIGN CONSIDERATIONS:

Hardware additions required:

- N-channel MOSFET (IRF540N or equivalent) on Arduino pin 3 (PWM)
- 220Ω gate resistor between pin 3 and MOSFET gate
- 1N4007 flyback diode across motor terminals (cathode to 9V+)
- 100μF smoothing capacitor across 9V battery terminals
- Momentary pushbutton on Arduino pin 2 with 10kΩ pull-down to GND
- 9V battery powering motor circuit; GND shared with Arduino GND

Software additions required:

- Task_Fan() added as 4th task in round-robin scheduler
- NUM_TASKS updated from 3 to 4
- Fan speed sequence: OFF(0) > LOW(26) > MEDIUM(38) > HIGH(64)

Note: Per project specification (HL.15), the original button cycle is OFF > LOW > MEDIUM > HIGH > MEDIUM > LOW > OFF. The simplified 4-step cycle (OFF > LOW > MEDIUM > HIGH > OFF) was implemented to improve button reliability. Update to 7-step cycle if required by evaluator.

END OF DOCUMENT

